EXP 15	Integrated Testing of the Library Management System

Aim:

To test the complete application end to end and confirm that the stock and the issue history are always correct.

Theory:

Integrated testing runs the business scenarios through the live forms so that every workflow is exercised. The function stockCheck() verifies for every book that Available Copies = Total Copies - copies not yet returned.

Procedure:

1. Run all scenarios in the test-case table through the live forms.
2. Create the function stockCheck() and execute it.
3. Compare the Book Catalogue, Issue Register and Dashboard with the expected values.

Program (stockCheck function):

```
void stockCheck()
{
    // verify: Available Copies = Total Copies - copies not yet returned
    for each bk in Books[ID != 0] sort by Book_ID
    {
        out = Book_Issues[Book == bk.ID && Issue_Status != "Returned"].count();
        if(bk.Total_Copies - out == bk.Available_Copies)
        {
            chk = "OK";
        }
        else
        {
            chk = "MISMATCH";
        }
        info bk.Book_Title + " | Total: " + bk.Total_Copies + " | Issued: " + out + " |
Available: " + bk.Available_Copies + " | " + chk;
    }
}
```

Output:

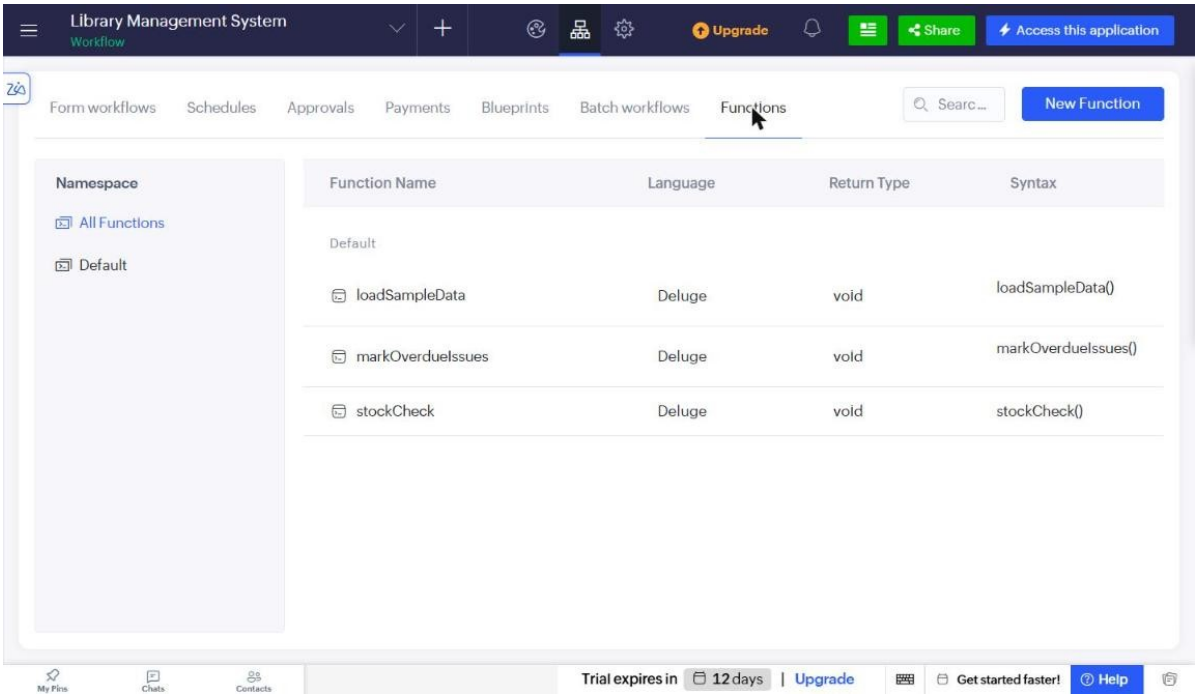


Fig. 15.1: Functions of the application

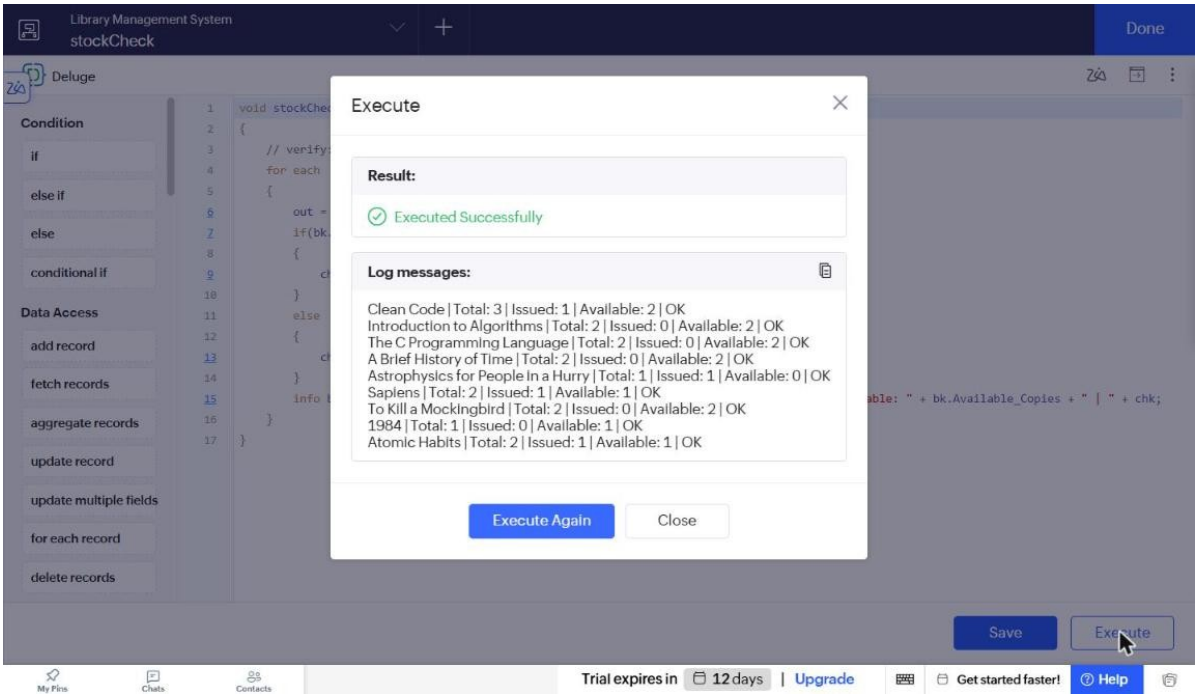


Fig. 15.2: stockCheck - every title OK

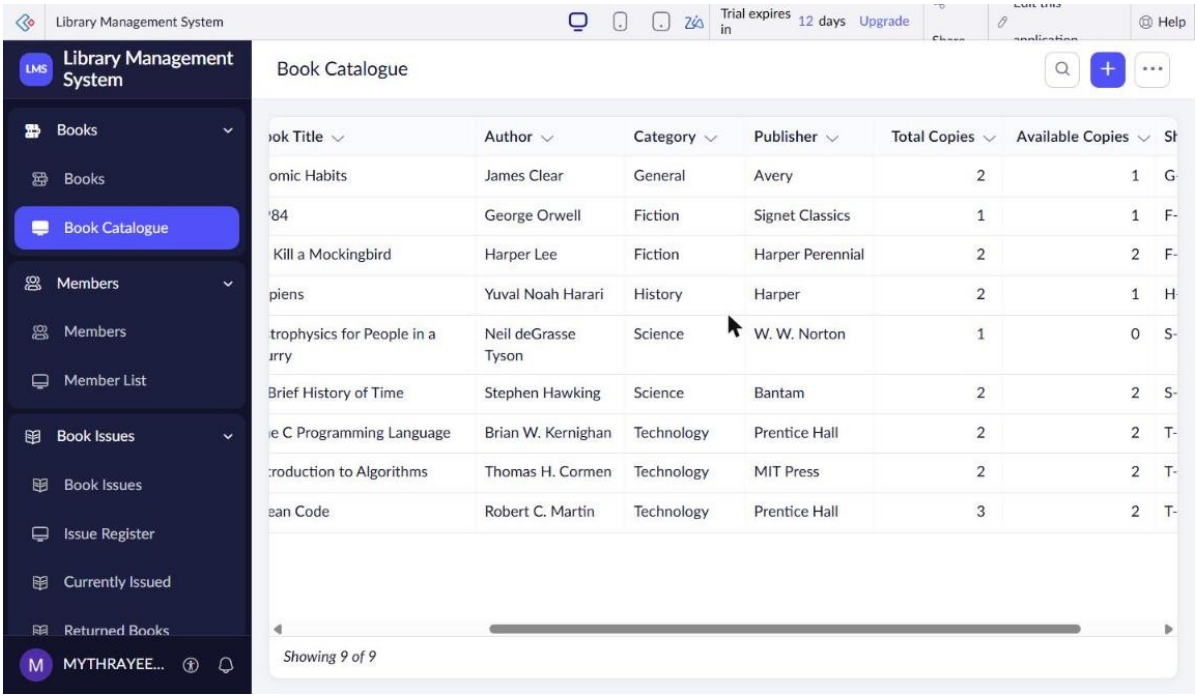


Fig. 15.3: Final stock in the Book Catalogue

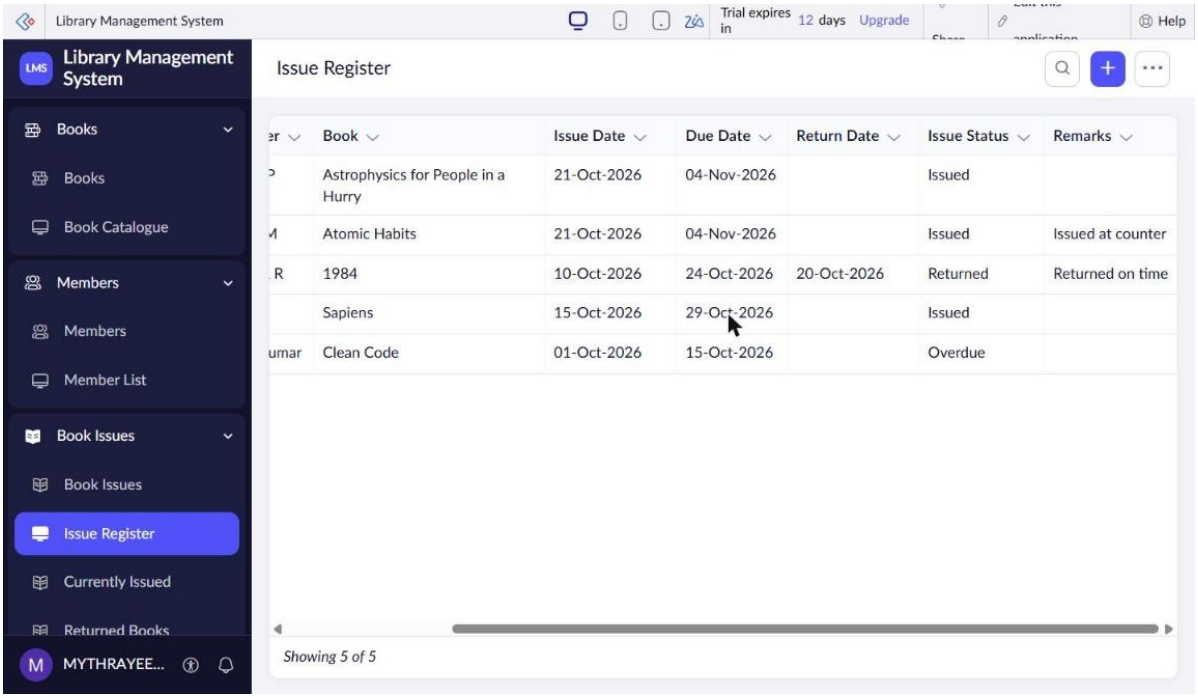


Fig. 15.4: Final Issue Register

Test cases:

No.	Test case	Expected	Actual	Status
1	Issue an available book (Divya M - Atomic Habits)	Saved, stock 2 to 1	Saved, stock 1	Pass
2	Issue a book with	Blocked	Blocked with	Pass

	no copies (Arun - Astrophysics)		message	
3	Issue to an inactive member (Rahul V)	Blocked	Blocked with message	Pass
4	Due date calculation (Issue 01-Oct-2026)	15-Oct-2026	15-Oct-2026	Pass
5	Return a book (Karthik - 1984)	Returned, stock 0 to 1	Returned, stock 1	Pass
6	Return the same book twice	Blocked	Blocked with message	Pass
7	Stock after multiple transactions	All titles OK	stockCheck: 9 of 9 OK	Pass
8	Overdue identification (due 15-Oct-2026)	Status Overdue	Overdue, shown in report and dashboard	Pass
9	History accuracy	All 5 issues kept with dates and status	Issue Register shows all 5	Pass
10	Book with Total Copies 0	Blocked	Blocked with message	Pass

Result:

All ten test cases passed. The Library Management System issues and returns books correctly, enforces the business rules and keeps the stock and history accurate.